Maximum Likelihood Estimation

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Normal lab: which μ explains the sample best?

The default population is $N(50, 16)$ with a sample of 10 observations. The variance is treated as known, and we compute $L(\mu) = \prod_{i=1}^{n} f(x_i; \mu, \sigma^2)$ for different candidates.

True μ: **50** | Variance σ²: **16** | Sample size n: **10**

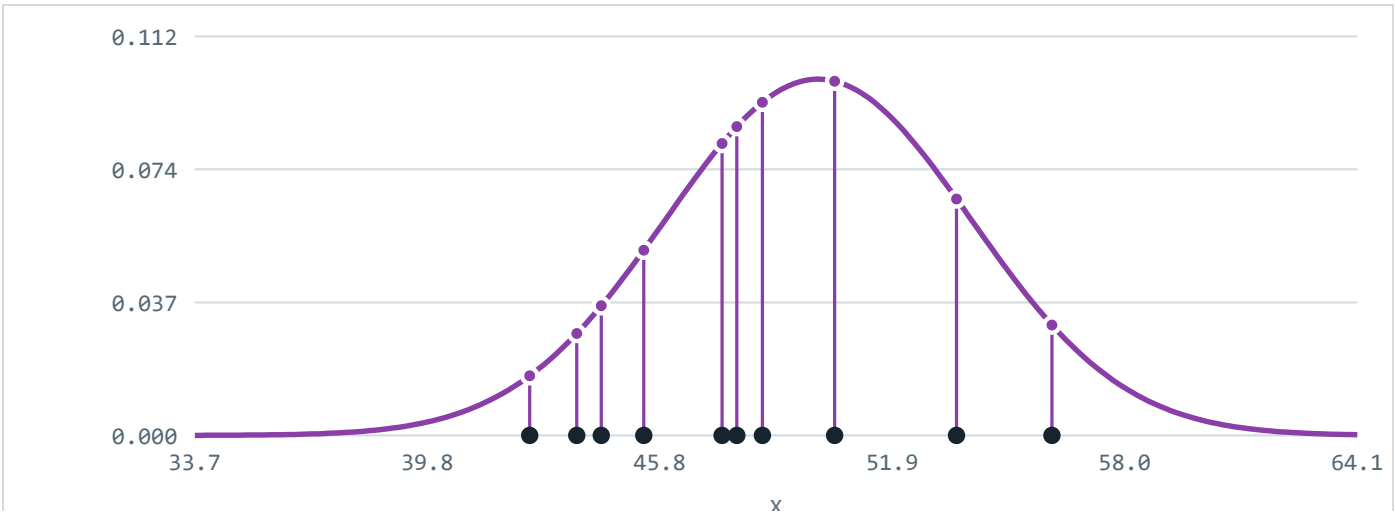
Draw new sample

☐ Plot ln L instead of L

OBSERVED SAMPLE

47.48 47.86 44.31 50.43 48.53 56.12 45.43 53.62 43.67 42.44

Sample mean **$\hat{\mu} = \bar{x} = 47.9887$** (true μ = 50). If σ² were also unknown, its MLE would be $\hat{\sigma}^2$ = **17.399** (true σ² = 16, s² = 19.332).

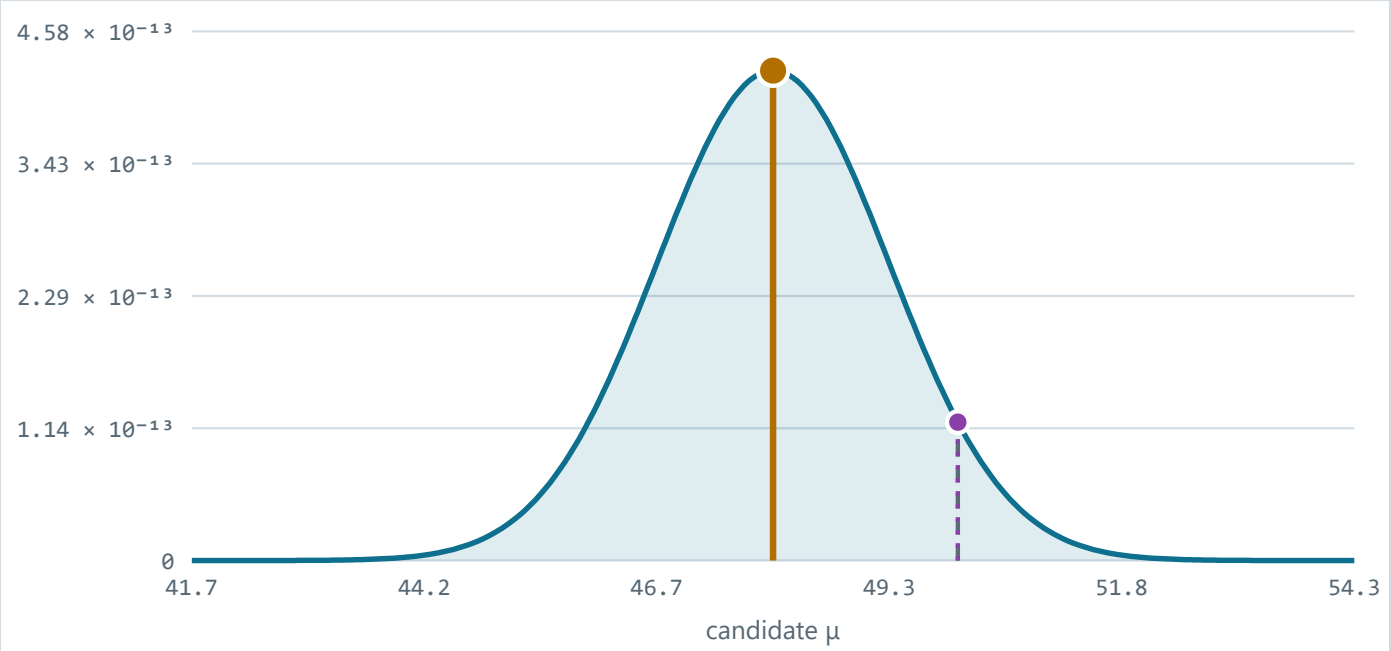


Candidate density $N(\hat{\mu}, \sigma^2)$ over the data. Each stem is $f(x_i; \hat{\mu})$. The likelihood is the product of the stem heights, so it is large when the curve sits over the data.

Your candidate μ̂: **50.00**

Jump to x̄

L(50.00) = **1.197 × 10⁻¹³**, ln L = **−29.754**. That is **28.2%** of the maximum L(x̄) = **4.238 × 10⁻¹³**.



μ̂ = x̄ (maximum) true μ your candidate

L(μ) for this sample. The peak is at μ̂ = x̄ = 47.9887, where L = **4.238 × 10⁻¹³**.

LIKELIHOOD AT CANDIDATE VALUES

CANDIDATE M	L	LN L	L / MAX
44	2.937 × 10⁻¹⁵	−33.461	0.69%
45	2.600 × 10⁻¹⁴	−31.281	6.13%
46	1.231 × 10⁻¹³	−29.725	29.1%
47	3.123 × 10⁻¹³	−28.795	73.7%
x̄ = 47.9887 MAX	**4.238 × 10⁻¹³**	**−28.489**	**100.0%**
48	4.238 × 10⁻¹³	−28.489	99.9%
49	3.079 × 10⁻¹³	−28.809	72.6%
50	1.197 × 10⁻¹³	−29.754	28.2%

	51	2.492×10^{-14}	−31.323	5.88%
	52	2.776×10^{-15}	−33.518	0.66%

The highlighted row is the sample mean. Every other $\hat{\mu}$ gives a smaller likelihood for this sample.